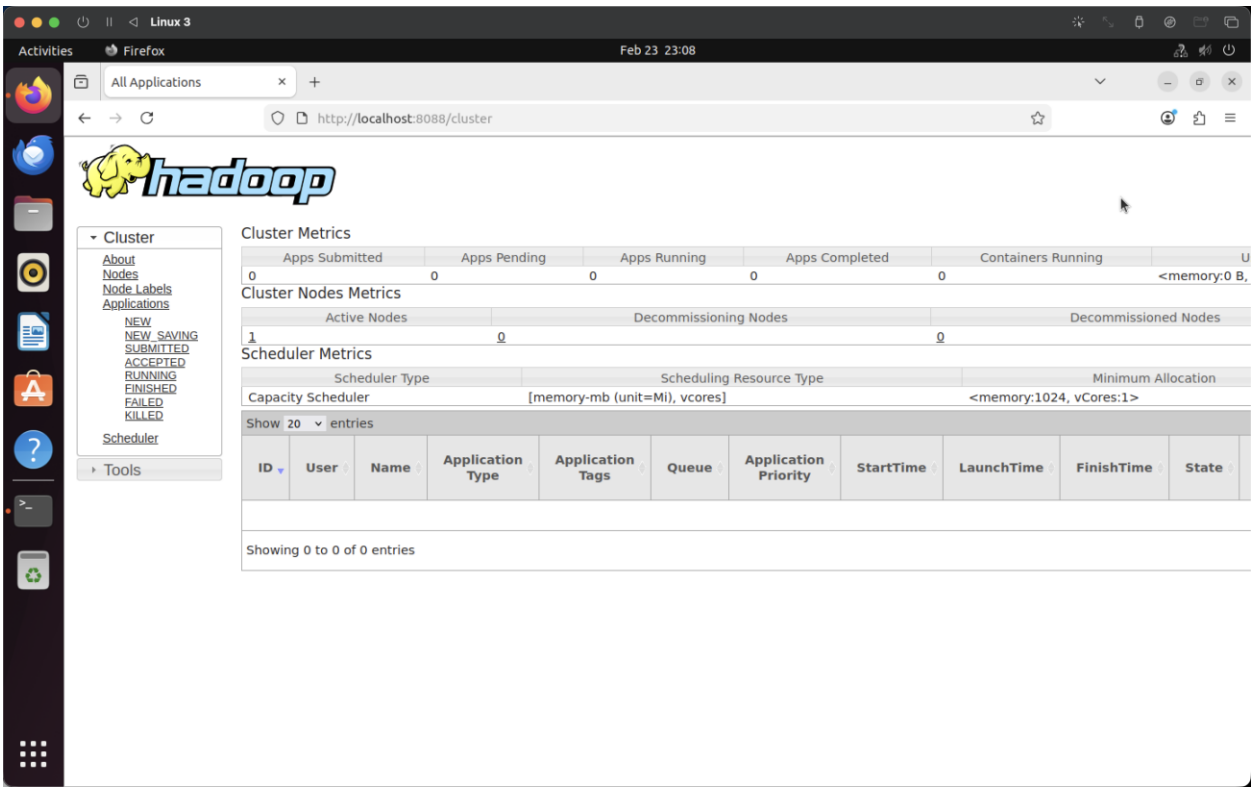
CS-4074

Name: Sara Imran **Student ID: S22107878**

Screenshots for Lab3

Task 1.1

Take a screenshot of the YARN ResourceManager Web UI (http://localhost:8088).



Task 6.1

Inverted Index Mapper:

```
#!/usr/bin/env python3
import sys
import os
filename = os.environ.get('mapreduce_map_input_file', '')
filename = os.path.basename(filename)
for line in sys.stdin:
    line = line.strip()
    words = line.split()

    for word in words:
        word = word.strip()
        if word == "":
            continue
        print(f"{word}\t{filename}"
```

Inverted Index Reducer:

```
#!/usr/bin/env python3
import sys
current_word = None
files = set()

for line in sys.stdin:
    line = line.strip()
    parts = line.split('\t')
    if len(parts) != 2:
        continue
word, filename = parts
```

```
    if word == current_word:
        files.add(filename)
    else:
        if current_word is not None:
        print(f"{current_word}\t{', '.join(sorted(files))}")
        current_word = word
        files = {filename}

if current_word is not None:
    print(f"{current_word}\t{', '.join(sorted(files))}")
```

Output:

```
analytics   doc2.txt
big    doc2.txt, doc3.txt
cluster       doc1.txt
computing         doc1.txt
data  doc2.txt, doc3.txt
distributed        doc1.txt
hadoop      doc1.txt, doc2.txt, doc3.txt
learning    doc2.txt
machine    doc2.txt
mapreduce        doc1.txt, doc3.txt
parallel      doc3.txt
processing        doc3.txt
spark        doc2.txt
```

Task 6.2

Mapper:

```
sara@sara-Standard-PC-Q35-ICH9-2009: ~
GNU nano 6.2                    /home/sara/temp_mapper.py
#!/usr/bin/env python3
import sys

for line in sys.stdin:
    line = line.strip()
    city, month, temp = line.split(",")
    city = city.strip()
    temp = int(temp.strip())

    print(f"{city}\t{temp}")
```

Reducer:

```
sara@sara-Standard-PC-Q35-ICH9-2009: ~
GNU nano 6.2                    /home/sara/temp_reducer.py
#!/usr/bin/env python3
import sys

current_city = None
temps = []

for line in sys.stdin:
    city, temp = line.strip().split("\t")
    temp = int(temp)
    if current_city == city:
        temps.append(temp)
    else:
         if current_city:
            print(f"{current_city} min={min(temps)}, max={max(temps)}")
         current_city = city
         temps = [temp]

if current_city:
     print(f"{current_city} min={min(temps)}, max={max(temps)}")
```

Output:

```
2026-03-12 15:50:20,331 INFO streaming.StreamJob: Output directory: /user/sar
ab3/temp_output
sara@sara-Standard-PC-Q35-ICH9-2009:~$ hdfs dfs -cat /user/sara/lab3/temp_out
/part-00000
Abha min=10, max=25
Dammam min=18, max=43
Jeddah min=25, max=40
Riyadh min=15, max=45
Tabuk min=8, max=38
sara@sara-Standard-PC-Q35-ICH9-2009:~$
```